Considerations regarding the project proposal "*Effects of biodiversity on ecological-economic ecosystem service provision*" based on data from the Jena Experiment

Several ecosystem functions have been identified that might be suitable to serve as proxy for different ecosystem services that shall be quantified in monetary terms. Here my considerations to those functions, whereas I will focus on the monetization of single functions and services, not on the overall focus of the paper.

1. Aboveground Biomass → Productivity

Straightforward: Derive economic value based on weight value of hay.

2. Nutrient and protein contents of plant biomass → Hay quality

It is true that nutrient and protein content (besides many other factors like the weather) affects hay prices. To avoid comprehensive surveys studies ("how much are farmers willing to pay for an increase of protein of X %") it might be possible to use data from agricultural statistics. Usually, the first cut of grass (around April-May) contains less protein than the final cut (around August-September). Comparing average prices of early- and late-cut hay on platforms like www.heupreis.de with the corresponding protein content might make it possible to extrapolate to derive economic value of hay per weight as function of its protein content. When doing this, one must keep in mind, that protein concentration might mainly depend on the sown species identity, not necessarily on the species richness.

3. Soil temperature → Cooling → Climate regulation (regional)

Studies focusing on the economic value of a cooler microclimate usually focus on "city ecosystems" (more trees and green infrastructure in a city leading to a cooler microclimate, using Willingness-to-Pay studies or others to determine the economic value). This probably cannot be transferred to agricultural ecosystems as there are usually no people around that would benefit from a cooler microclimate (expect the farmers that are working on the field). Also, although the cooling effect of biodiversity might be significant in the soil, I'm not sure if this has a recognizable effect on the air (something that humans would actually notice). Probably the effect is rather indirect (cooler soil leading to reduced stress of plants, leading to higher yield). In this case, double counting should be avoided (as the economic value of soil climate regulation would already be reflected in the yield).

4. SOC → Carbon sequestration → Climate regulation (global)

Use biomass conversion factors to determine the CO2-equivalent stored in the soil (check the soil properties to apply the corresponding conversion factors) and use data from the UBA (social cost of pollution) or compliance markets (ETS / Corsia) for monetization (sensitivity analysis?)

5. **Soil N + Soil P after growing season // N/P in soil solution → Water pollution, nutrient leaching → Water regulation/quality**

Could be done with a similar approach as in Scherzinger et al., 2024. The soil characteristics of the area of the Jena experiment shall be checked. If the soil has a high water holding capacity and a high enzymatic activity, N/P might be completely denitrified before leaching to surface water / groundwater. In such case there would be no or few social cost (if working with the UBA values for N/P leaching into groundwater).

6. **Leaf area index + (Microbial biomass +) soil water content + soil infiltration rate → above- and belowground water holding capacity, velocity of soil water uptake à flood protection**

Interesting. Probably there is no strong effect of biodiversity on the water holding capacity of the soil / infiltration rate (as this is mainly based on the general properties of the soil – still it would be good to check). But there might be an effect on leaf area index. I remember there was a flood in Jena that also affected the experiment. It might be possible to derive an economic value from the water retention based on the number of households / the value of the infrastructure in Jena that would be / was affected by such an event.

7. **Aboveground biomass stability → Yield stability → Yield stability (insurance)**

I remember there was a significant effect of biodiversity on stability. This could be monetized with a similar model as in Scherzinger et al., 2024. Also, a 2d-model could be used (instead of the 3d model). I would recommend a model like this:
RP(b) = A * exp(-kb) with A>0 to model risk premium (RB) as function of biodiversity (b).

8. **Pollination → Pollination rate → Pollination**

Careful. Might already be reflected in the biomass productivity. I remember that some newer studies completely refrain from including supporting services to avoid double counting (moving away from the traditional concept of 4 ecosystem service categories as shown in the Millennium Ecosystem Assessment), but only monetizing the final services that provide the actual value to humans (production, regulating, culture).

9. **Flower cover → Aesthetic value**

Not sure if there is a recognizable effect.

10. **Per capita effects of aboveground herbivores on plants (via energy fluxes, Barnes et al 2020 Sci Adv) → Aboveground pest control**

Here there might be a similar problem as for pollination as pest control is probably already reflected in the yield. Still, if it is a common practice that farmers actively do pest control in grassland, it might be possible to derive a “avoided cost” value (how much money for pesticides can be saved).

11. **Per capita effects of belowground herbivores on plants (via energy fluxes) à Belowground pest control**

See point 10.

12. **Leaf area affected by pathogens (inverse) à Pathogen control**

See point 10.

13. **Soil biodiversity (topic of Marius' thesis)**

Economic value of soil biodiversity could be directly derived from Scherzinger et al., 2024.

14. **Soil health index (topic of Marius' thesis): Microbial biomass + Decomposition rate + enzymatic activity + …**

There also might be a stability effect?